

A novel Bayesian multilevel regression approach to the reconstruction of an eastern Mediterranean temperature record for the last 10,000 years

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Abstract

Climate reconstructions derived from proxy records for individual sites often fail to incorporate existing regional information, which may help to determine uncertainties and express variability within specific reconstructions or commonalities between datasets. Such an understanding is crucial when examining past human-environment interactions. Taking the eastern Mediterranean as our case study, we present here a new air temperature model for the last 10,000 years that utilises data from 33 previously published proxy-based independent reconstructions, using a novel fully Bayesian approach that applies multilevel regression models of individual temperature datasets grouped together into 300 year long consecutive sub-intervals. A Bayesian multilevel approach allows the model to share information between each regression model from the individual datasets and the 300 year grouped regression models. The results demonstrate commonalities between individual datasets derived from different sources, and embed the uncertainties within the model. Our results establish that the eastern Mediterranean region was consistently warmer than the 20th century, except for two short intervals at the end of the Early Holocene (between 8400 and 8250 yrs BP) and the start of the mid-Holocene (between 7800 and 7650 yrs BP). We also identify changes within temperature associated with both the 8.2 ka and 4.2 ka BP events, however our findings identify regional warming in the eastern Mediterranean, rather than cooling often associated with the 4.2 ka BP event. Our results are comparable with previous large scale hemispheric reconstructions, demonstrating that our model is a robust candidate for temperature reconstructions within a confined region, which can range from mesoscale up to macroscale.

Keywords

Bayesian hierarchical model, eastern Mediterranean, Gibbs sampling, MCMC, paleoclimate, the Holocene

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Introduction

Understanding temperature variability at a regional scale is critical as it informs discussions across a range of disciplines, from palaeoecology to past human-environment interactions and, by inference, can contribute to discussions of future climate scenarios. Paleoclimate reconstructions based on proxy data from a particular location are subject to a variety of uncertainties, which can be summarised as those resulting from data sampling, analytical techniques, age-depth models and the nature of the proxy data themselves. Furthermore, climatic variable reconstructions solely based on data from a specific location do not incorporate data from extant studies other than as a basis for comparative analysis, as such, they fail to incorporate a wealth of existing prior knowledge. As such, the complex and varied nature of temperature data available from climate proxies for the eastern Mediterranean region can make it challenging to comprehend long-term regional climatic trends (Roberts et al., 2010). Therefore, the existing temperature datasets, even for micro-to mesoscale regions such as the eastern Mediterranean, may display noticeably differing trends or levels of variability (Figure 1).

There have been earlier reconstructions of global temperatures based on non-Bayesian methods (Kaufman et al., 2020; Marcott et al., 2013; Snyder, 2016b). However, Bayesian techniques are

more flexible than non-Bayesian approaches in terms of reflecting and assimilating prior information into the model as well as representing the uncertainty level of the results in terms of probability, and their results are more comprehensible. There have also been reconstructions of the climate field through Bayesian estimation of a hidden Markov model, which links instrumental and proxy data (most notably, Tingley and Huybers, 2010 and its extensions for temperature reconstructions Zhang et al. (2018), Tingley and Huybers (2013); and for precipitation reconstructions Gómez-Navarro et al., 2015; Talento et al., 2019). The model suggested by Tingley and Huybers (2010) is remarkable in terms of the complexity of temporal and spatial reconstruction, but it is dependent on the calibration of proxy data to instrumental

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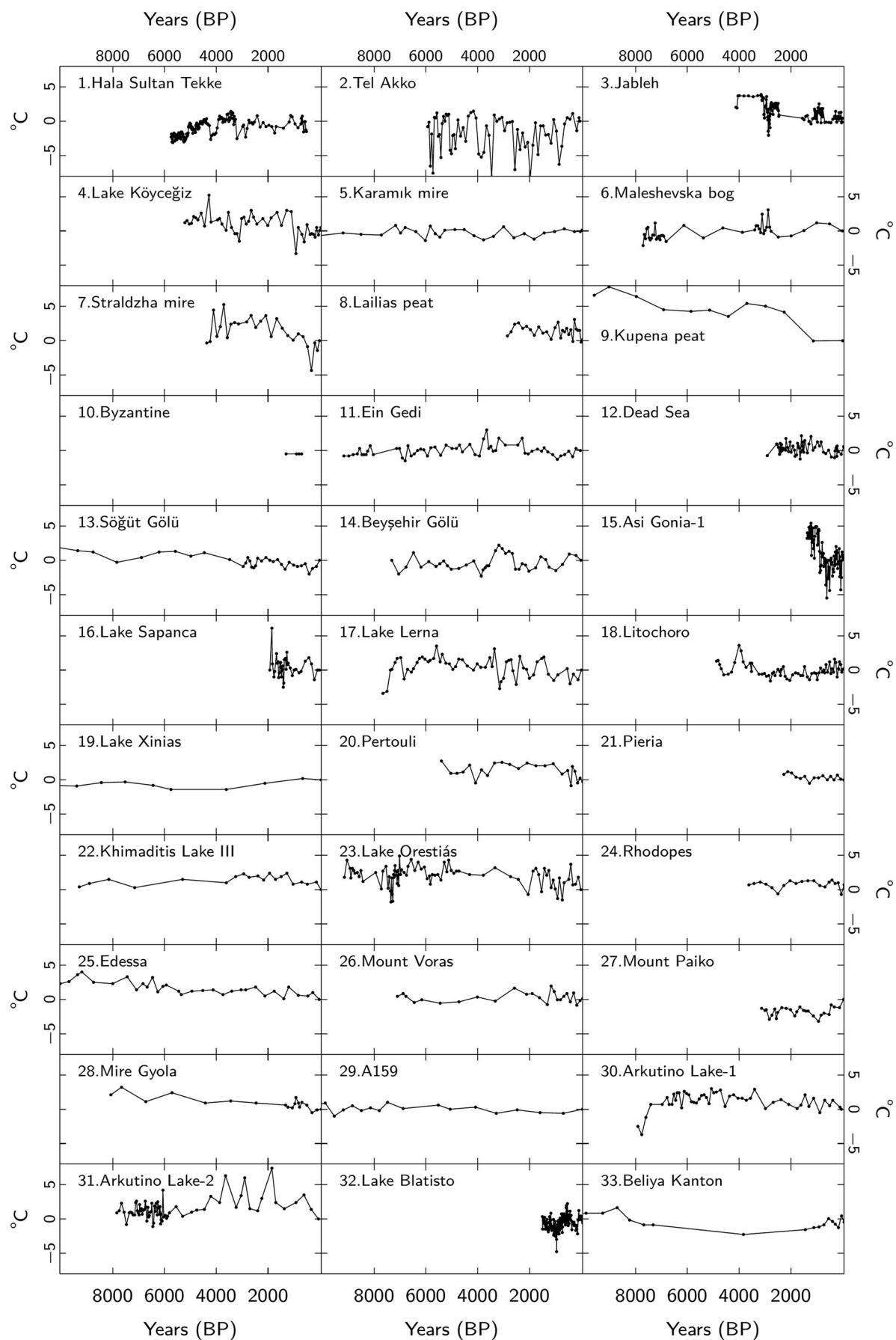


Figure 1. Time series plots of temperature anomalies in $^{\circ}\text{C}$, defined as departure from the 20th century average (see Data section) for the datasets given in Table S1, available online.



Figure 2. The location map of the data given in Table S1, available online. The map was prepared using GMT (Wessel et al., 2019).

observations. Furthermore, because the model is based on an autoregressive process, it requires data that are evenly spaced.

However, available temperature data have a low resolution, and the data points are highly scattered over time. Therefore, we applied a fully Bayesian multilevel regression model (comparable to the sea surface temperature model of Snyder, 2016a) that can incorporate multiple existing temperature datasets for a region with any resolution or spacing. Our model can explicitly optimise and report the information shared between each dataset (see Gelman and Hill, 2006) even when they display noticeably differing trends or degrees of variability (Figure 1).

Using the Bayesian multilevel regression model of this study allows us to take advantage of existing data available in published environmental reconstructions by identifying commonalities within disparate datasets to produce an average temperature curve for a single large and geographically coherent region. The selection of the eastern Mediterranean is advantageous as much academic discussion has focused on its human settlement history (Lawrence et al., 2021), landscape change (Welch and Marks, 2014), past human adaptations in response to climatic changes (Roffet-Salque et al., 2018) and the influence of abrupt Holocene climatic events (Ön et al., 2021; Ur, 2015; Weiss, 2015); with a wealth of archaeological evidence, palaeoclimatic samples and

competing hypothesis concerning the human and environmental evolution of the region (Akçer-Ön et al., 2020; Grove and Rackham, 2003).

We generate annual mean temperature deviations from 1950 CE (0 yrs BP) back to 9900 yrs BP¹ for every 150 years. The corresponding temperature values for each point are found by taking averages of data spanning 150 years on either side of these points (i.e. 300 year blocks). This interval size was selected considering both the variability and resolution of the data. For each interval, we used temperature data from a physical climate model – TRaCE21ka (Liu et al., 2014), as a prior. The results of a multilevel Bayesian regression approach realistically acknowledge the complexity and variability of the original datasets. As Walker et al. (2007: 369) recognise, ‘In many situations, Bayesian inferences can be regarded as “more honest” in the manner that they simultaneously account for multiple uncertainties’.

Our study presents two original elements: the first is the model itself, and the second is the results from the application of that model to a group of datasets from the eastern Mediterranean. In this study, we identify 33 existent proxy based temperature reconstructions from the eastern Mediterranean (Figure 2). We propose a model which is based on Bayesian hierarchical linear regressions on individual data within groups over evenly spaced smaller

intervals (300 years) for the last 10,000 yrs BP. As a result, we present a single composite temperature reconstruction curve for the eastern Mediterranean region that also expresses Bayesian credible uncertainties. The results presented here are based on the current state of the research available and have the potential to contribute to several discussions in ecology and human-environment interactions (Dee et al., 2020). Our model can accommodate other palaeoclimatic studies and new data as and when they are published, recognising the dynamic nature of this research field (a detailed methodology and criteria for data inclusion are provided within the Data section).

Materials and methods

This section details the data and methods used within this study; emphasis is placed on the data sources, criteria for inclusion and the novel methodology developed to ensure it can be replicated in subsequent studies.

Data

We do not restrict our analysis to any specific form of temperature proxies (e.g. pollen records), instead we incorporate different types of proxy temperature reconstructions. However, within the eastern Mediterranean geographical region, the available proxy reconstructions that fulfil our criteria for inclusion are those predominantly based on pollen. In applying our Bayesian model via selected priors, we needed to make certain decisions, one decision is the inclusion criteria for datasets and the other is setting the duration of the time intervals. We therefore propose the following criteria for each climatic reconstruction's inclusion:

1. Datasets must lie within a defined region. Within this study we focused on the eastern section of the Mediterranean bio-climatic zone which we decided to delimit by the Balkan mountain range to the north, the Pindus range to the west, the Levant to the east and the North African coast (Figure 2).
2. Datasets should be publicly available and include air temperature reconstruction with a proper age model (for details on the construction of the age models, see the relevant references in Table S2, available online).
3. Each dataset requires at least one data point within the 20th century.

The available data are concentrated in particular regions, such as the Balkan Peninsula, whilst others offer few sources currently for example, the North African coast. However, as new data become available from the eastern Mediterranean, they can be assimilated in our reconstruction.

Within our analyses, we take into account the temperature deviation of each record from its 20th century level, we select locations (Table S1, available online) which have at least one data point in the 20th century. If, for a data location, there is only a single measurement point in the 20th century, then we consider it to be the average value of the century. However, if multiple points exist, we simply take the arithmetic average of these multiple points as representative of the 20th century. Finally, we find the deviation of each dataset by subtracting the 20th century average from each data point in the set.

Having applied our inclusion criteria, 33 existent datasets were identified. However, the limited number of data points within each dataset led us to set time intervals at 300 years; if the available datasets in other regions, for example, contained higher resolution records, shorter intervals may be considered optimal. If we were to have more closely defined time intervals (e.g. a decade) there would have been a considerable increase in uncertainty and

consequent loss in accuracy. If, however, we were to set the time intervals at a longer period (e.g. a millennium) then there would be such a loss of variability within the results and too few data points in the results to make it difficult to identify any meaningful trends.

Our model is based on multilevel linear regressions for each interval with the assumption of finding the temperature candidate for the interval through a linear trend, but more importantly we assume a commonality in the evolution of each individual dataset (reconstructed series) and the data as a whole. The results are then used to predict the midpoints as representative of temperature anomaly for each interval via posterior predictive distribution (PPD). Hyperparameters for the prior distributions of the model for each interval are retrieved from the region enclosed between 25°–30°E and 35°–40°N of a numerical global coupled global general circulation model called TRaCE21ka, through an interface called PaleoView (Fordham et al., 2017). While our belief in the prior is neither strong nor moderate, we implement the TRaCE21ka data into the prior of the model as being moderately informative. Setting TRaCE21ka as strong or weak priors, or even as very weak priors with slope and intercept parameters that are equal to zero makes little substantial difference to the point estimations (i.e. the median values), but does affect the uncertainty intervals (Figure S265, available online).

Model

We start from 150 yrs BP and apply the following described algorithm for all points in the set $T = \{150, 300, \dots, 9900\}$ yrs BP. For each $t \in T$, we select a window of length 300 years that equally disperses on both sides (± 150 years) around t . From now on, unless specified, we will call the selected window as the interval and the selected t as the midpoint (Figure S266, available online).

While we present our model as a Kruschke diagram at Figure 3, the model description is as follows;

$$y_{i|j} \sim t_v(\mu_{i|j}, \sigma^2) \quad (1)$$

where,

$$\mu_{i|j} = \beta_{0,j} + \beta_{1,j}x_{i|j}. \quad (2)$$

First level priors for the model are,

$$\sigma \sim \text{half - Cauchy}(A_\sigma, B_\sigma) \quad (3)$$

$$(v-1) \sim \text{Expon}(L) \quad (4)$$

$$\beta_{0,j} \sim N(\mu_0, \sigma_0^2) \quad (5)$$

$$\beta_{1,j} \sim N(\mu_1, \sigma_1^2) \quad (6)$$

$$x_{i|j} \sim t_{v=1}(\bar{x}_{i|j}, S_x^2) \quad (7)$$

and, second level priors are,

$$\mu_0 \sim N(M_0, S_0^2) \quad (8)$$

$$\mu_1 \sim N(M_1, S_1^2) \quad (9)$$

$$\sigma_0 \sim \text{Inv - gamma}(A_0, B_0) \quad (10)$$

$$\sigma_1 \sim \text{Inv - gamma}(A_1, B_1) \quad (11)$$

Here, the subscript j represents the data source, and i represents the total observations within each window. The subscript notation $i|j$ means the i th observation in the whole data that lies within the j th individual.

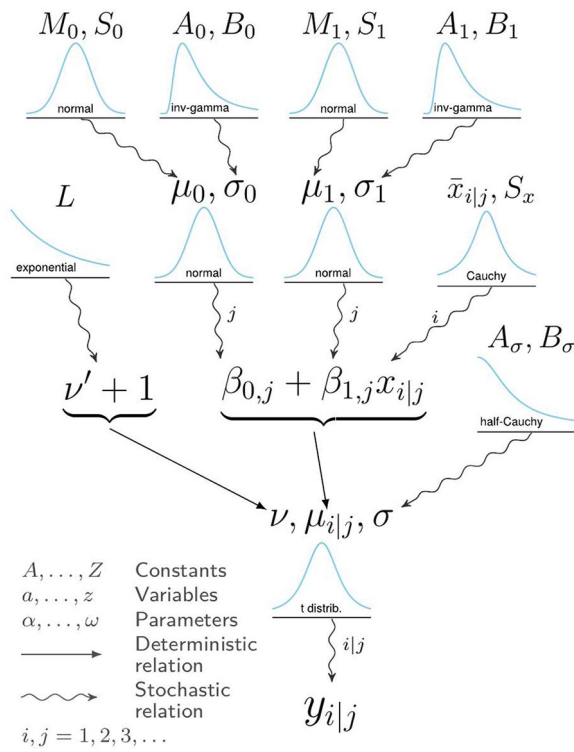


Figure 3. Kruschke style diagram (Kruschke, 2015) of the hierarchical linear regression model for all temperature anomaly values. The letters next to the curly arrows indicate at which dimension the priors are defined.

In our linear regression model, each i th observation of j th dataset, $y_{i|j}$, is a random value described by a t -distribution (equation (1)). Selection of the t -distribution allows the regression to be immune to outliers. Its location parameter $\mu_{i|j}$ is defined by $\beta_{0,j} + \beta_{1,j}x_{i|j}$ (equation (2)). The normality (or call it the degrees of freedom) parameter, ν , is given a broad exponential prior (equation (4)). Following Kruschke (2015: 462), the hyperparameter L of equation (4) is selected as $1/29$. By doing this, the mean of ν is selected as 30, resulting in an equal allocation of weight to values above and below 30. The distribution of the scale parameter of the t -distribution, σ , is selected to be half-Cauchy (equation (3)) as offered by Gelman (2006). The location parameter is naturally selected to be $A_\sigma = 0$ and the scale parameter $B_\sigma = 3$.

Every individual dataset in an interval has its own intercept and slope parameter, namely $\beta_{0,j}$ and $\beta_{1,j}$. They are both estimated through normal distributions $\beta_{0,j} \sim N(\mu_0, \sigma_0^2)$ (equation (5)) and $\beta_{1,j} \sim N(\mu_1, \sigma_1^2)$ (equation (6)). Here, μ_0 and μ_1 are defined through normal distributions with mean values M_0 and M_1 and standard deviation values S_0 and S_1 , respectively. The hyperparameters M_0 and M_1 at equations (8) and (9) are pulled from the TRaCE21ka transient climate model data through PaleoView (Fordham et al., 2017). For each interval, the intercept (M_0) and the slope (M_1) of the TRaCE21ka is established using the values of the endpoints of TRaCE21ka for the same interval taken from PaleoView. However, our prior beliefs in μ_0 and μ_1 are different. Since our belief in the intercept is relatively weaker than the slope and selecting a strong prior for the intercept can pose spurious slope values, we select $S_0^2 = 3^2$, and, we select a stronger prior for the slope, which is $S_1^2 = 2^2$. This selection is determined in balance for the intervals with relatively low and

high numbers of data points. Furthermore, we use uninformative priors for σ_0 and σ_1 , where the parameters of the inverse-gamma distributions (A_0, B_0, A_1, B_1) are selected as 0.01 (equations (10) and (11)). Our model is applied on the standardised age and temperature data (Kruschke, 2015) within each interval. This transformation is undertaken to avoid Gibbs sampling getting stuck, and consequently enables convergence to the desired posterior distributions more efficiently. We further implement a small amount of age uncertainty for each data point. This is established through Cauchy distributions (equation (7)), where for each datum the location parameter is selected as the age point for the datum itself and the scale parameter is selected as 0.1 within the standardised scale.

Posterior distributions of the model are handled by Gibbs sampling via JAGS (Plummer, 2003) through runjags (Denwood, 2016) in R (R Core Team, 2020). The Gibbs sampling process in JAGS is executed after 10,000 adaptation steps and 10,000 burn-in steps are discarded. For each interval, three Markov chain Monte Carlo (MCMC) chains are run, each with 150,000 iterations. Convergence for each interval is inspected via trace plots, autocorrelation plots, density plots and Gelman-Rubin statistic plots for each chain (Figures S1, S5, . . . , S261, available online). If a convergence problem is suspected, then 600,000 iterations are run for each chain. Convergence diagnostics plots of MCMC runs are handled through functions given by Kruschke (2015). Point and interval estimates are made through the median and the highest density interval (HDI) of the PPD of each interval (Figures S2, S3, S4, S6, S7, S8, . . . , S262, S263 and S264, available online), where PPD is found through sampling from $p(y | \mu_{i|j}, \sigma^2)$.

All the code and data used within this study are freely accessible, with reanalysis intended as new reconstructions become available.

Results and discussion

The results from our temperature model (Figure 4; Table S3, available online) identify that the median values of the PPDs are higher than the 20th century normal in the period between 7500 and 1500 yrs BP. In the period before 7500 yrs BP, variability of the record increases, with two incidences where median reconstructed temperature values fall below the 20th century normal. Proxy based temperature reconstructions of larger regions such as the extratropical Northern Hemisphere (30°–90°N) by Marcott et al. (2013) and of the mid-latitudes (30°–60°N) by Kaufman et al. (2020) are comparable with our model as they all demonstrate average temperatures are consistently above the 20th century normal in the period 7500–1000 yrs BP (Figure 5). For the last millennium the Marcott et al. (2013) data show a gradual and then rapid decrease in temperature below the 20th century normal, whereas our results show a more stable temperature just above the 20th century normal comparable with the reconstruction of Kaufman et al. (2020).

Uncertainty intervals as expressed as the 89% HDI of the PPD, are narrower after 7500 yrs BP than before this date, indicating greater uncertainty in the early record. Our results (Figure 4) demonstrate that there are intervals where uncertainties are high, especially before 9000 yrs BP (see Figure S267, available online), reflecting either the existence of few data points in the whole dataset or a small number of individual data points. This explains the greater variability and uncertainty within our model compared to that of Marcott et al. (2013) and Kaufman et al. (2020) who use larger areas and therefore have more data for this early period (9000–7500 yrs BP). Furthermore, intervals where there are few available data points and where trends and intercepts correspond poorly with the trend within the TRaCE21ka physical model,

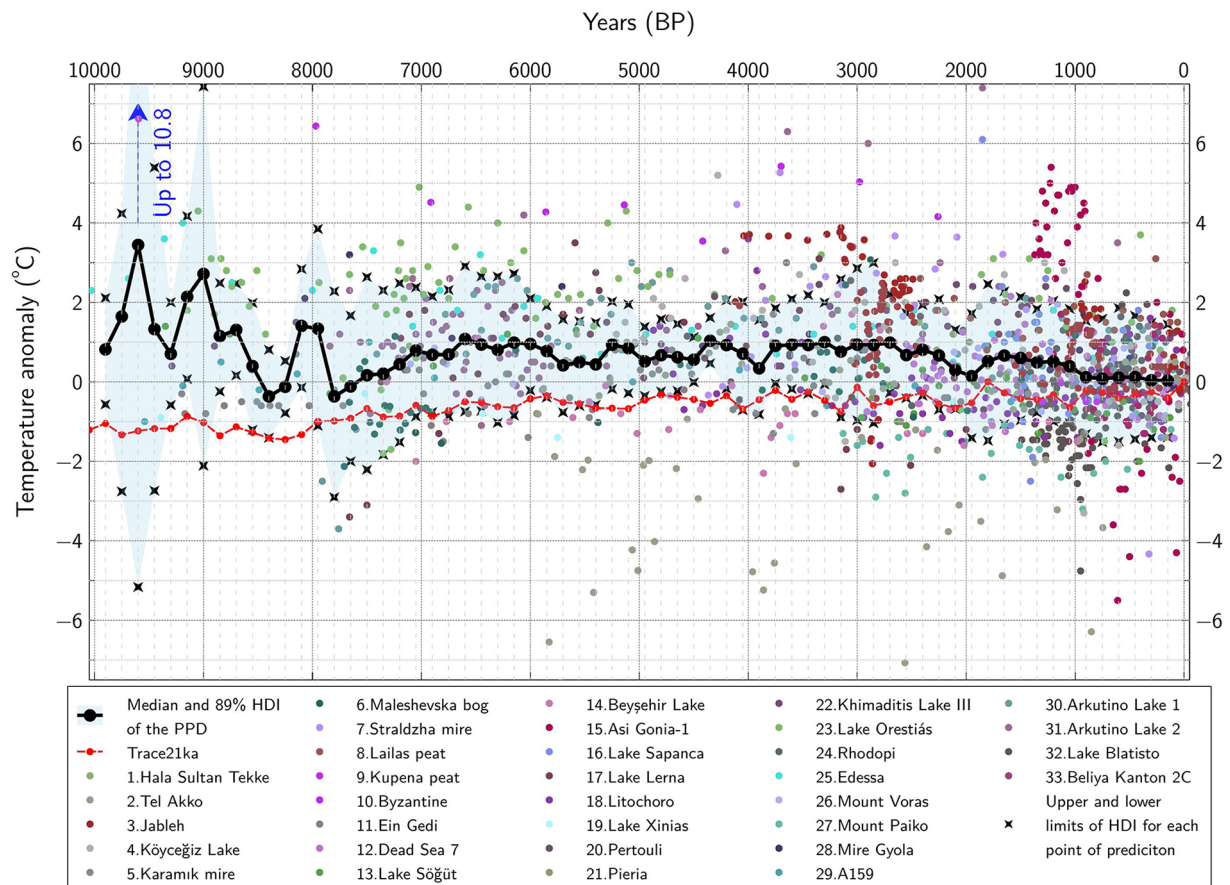


Figure 4. The result of the whole analysis. The black thick points and the sky blue shading represent the median for each sampling point (which are 150,300,...,9900 yrs BP) and the 89% HDI of the posterior predictive distribution, respectively. The endpoints of the HDI for the midpoints, shown with a cross, of each interval are linearly connected. The red curve shows the TRaCE21ka data (Fordham et al., 2017) used through parameter assignments of prior distributions.

Note. Please refer to the online version of the article to view this figure in color.

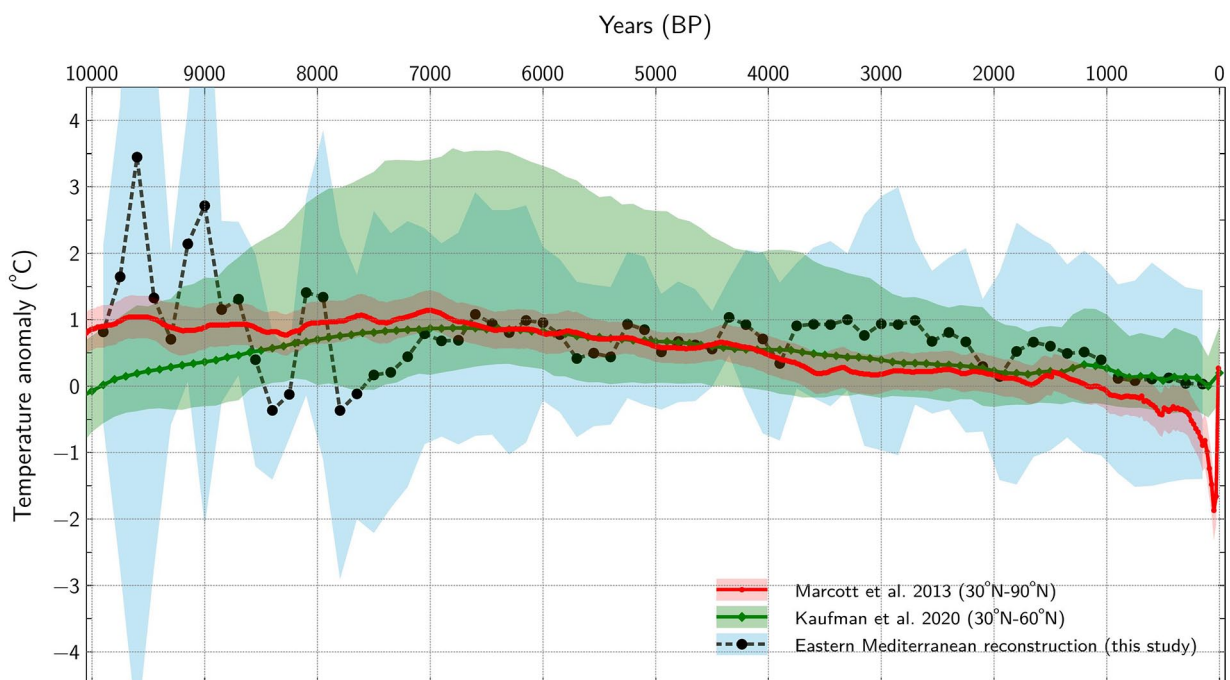


Figure 5. Comparison of our results with the temperature anomaly reconstructions of Marcott et al. (2013) for 30°–90°N latitudes and Kaufman et al. (2020) for 30°–60°N. While the temperature reconstruction we offer gives the temperature anomaly according to the 20th century average, reconstructions of Marcott et al. (2013) and Kaufman et al. (2020) are the temperature anomalies according to the average of 1961–1990 CE and according to the 19th century average, respectively.

results in a substantial increase in uncertainty (e.g. see Figure S254, available online). However, if, at these time intervals or for the entire analyses, the prior belief in the TRaCE21ka model increases, in order to decrease uncertainty, this would devalue the observed proxy data against the physical model. For each midpoint after 8500 yrs BP, the corresponding 89% HDI includes the zero point (0°C), except for the 4350 yrs BP midpoint. This identifies that Holocene temperatures were not statistically different to those of the 20th century, except, probably, at the 4350 yrs BP midpoint.

For exactly the same intervals, simple arithmetic means were calculated and assigned as the representative value for each interval. These outcomes were then compared to our Bayesian reconstruction (Figure S268, available online). With the exception of the interval between 6000 and 4050 yrs BP, the moving average results not only fall within the credible intervals but also agree quite well with the median values of our reconstruction (see the yellow curve in Figure S268a, available online). Closer inspection of the data reveals that Hala Sultan Tekke (Kaniewski et al., 2020) and Tel Akko (Kaniewski et al., 2013) temperature reconstruction data are almost consistently below 0°C (see Figure 1, Unit 1 and Unit 2 in Figures S107, S108, S111, S112, S115, S116, . . . , S155 and S156, available online) according to our method of finding the temperature anomaly (see Data section). If these two datasets are removed and the moving arithmetic averages of the data examined, the results lie within credible intervals and are similar to our results for all intervals (see the magenta curve Figure S268a, available online and the bar charts given under Figure S268b, available online). This can be seen as evidence of the robustness of our model to outliers. Therefore, it can be said that meta-analyses based on Bayesian statistics that describe the data through heavy-tailed distributions, such as the t-distribution in our case, can provide better solutions than noisy data that are based on a single site.

Two notable abrupt climate events and their possible impact on the environment and ancient cultures of the eastern Mediterranean are currently being widely discussed, commonly known as the 8.2ka BP event (Flohr et al., 2016) and the 4.2ka BP event (Marshall, 2022; Ön et al., 2021). The 8.2ka BP event is defined as a period of cooling occurring between 8250–8090 yrs BP (Rasmussen et al., 2014). Even though the number of different individual datasets and data points that constitute our model parameters of the 8250 yrs BP midpoint are low, this interval still exhibits low uncertainty. The 8250 yrs BP point reveals almost the lowest median temperature value within the whole series, however, the HDIs do not exclude the zero point. These findings add important support to the call by Berger and Guilaine (2009) for further evidence for the 8.2ka BP event within the Mediterranean region.

The chronological span of 4.2ka BP event was precisely defined as 4200–3900 yrs BP when first proposed by Weiss et al. (1993), although subsequent publications have applied less precise chronological limits (Zanchetta et al., 2019). While the 4.2ka BP event is primarily conceived of as a hydrological and aeolian event (Weiss et al., 1993), subsequent publications (Weiss, 2015) state that colder temperatures are to be associated with this phenomenon. It is therefore interesting to note that we see no evidence for a cooling during the 4200–3900 yrs BP interval in our model. However, within our model the only point that we identify as statistically different from the 20th century after 7500 yrs BP is the 4350 yrs BP midpoint, which incorporates data from 4500 to 4200 yrs BP; the HDI indicates that temperatures were at least 0.5°C warmer than the 20th century, with a median of 1°C warmer. As noted by Bini et al. (2019) there is uncertainty in the synchronicity and timing of the 4.2ka BP event, even when accommodating uncertainties within dating approaches. The period is typically considered to be one of cooling across the Mediterranean. However, the results from our study demonstrate above normal warming, corroborating the findings of Kaniewski et al. (2018), who

also note that warming is consistent with other sites in the eastern Mediterranean.

Conclusions

We propose here a novel Bayesian multilevel regression model for regional temperature reconstruction for the last 10,000 years, using multiple independent datasets derived from different climate proxy sources. When applying our model, the analysis or the results can be expressed at multiple resolutions, subject to the availability of data or the nature of the intended research question. Our results demonstrate:

1. The median of the temperature reconstruction results is almost always positive, that is, above the 20th century normal, with few exceptions.
2. The uncertainty intervals always include zero, with two exceptions (9150 and 4350 yrs BP), that is, according to our results it is statistically possible that temperatures for this period were no different from the 20th century.
3. The magnitude of the uncertainty intervals is higher before 9ka BP as a result of limitations in the dataset.
4. Even though we use TRaCE21ka temperature data as a moderate prior within our Bayesian model, the results of our regional temperature reconstruction shows a markedly different pattern to that presented by this global physical climate model. However, our results show similarity to the Northern Hemisphere statistical temperature reconstructions, which are based on different datasets to our own, proposed by Marcott et al. (2013) and Kaufman et al. (2020).
5. As a result of the nature and design of the method presented, inclusion of additional datasets as they become available will likely reduce the uncertainties within the module, and therefore, the approach will likely result in more robust reconstruction results in the future, with easy applicability and reanalysis.
6. The focus of this reconstruction is the eastern Mediterranean; however, the approach developed can be readily applied within any region or climatic zone. Therefore, new future regional studies can be undertaken by applying this model to regionally available datasets. Similarly, the method deployed within this study could in principle be extended to precipitation reconstructions, however, consideration would be needed of regional precipitation distribution characteristics, probability distributions to be deployed in the model, seasonal-annual regime stationarity and potential input parameters.
7. The development of mesoscale to macroscale regional temperature reconstructions offers considerable opportunities for reassessment of impacts of abrupt climatic events in particular regions, such as the 8.2ka BP and 4.2ka BP events in the eastern Mediterranean. However, the approach also offers opportunities for reassessment of regional climate-landscape-human interactions. This is crucial for understanding sociocultural development of societies, offering important opportunities for assessing and reassessing the relationships and interactions between early human communities, ecology and climate.

Acknowledgements

All the graphs in the main text are plotted through an open source scientific plotting package, Veusz, version 3.4 (<https://veusz.github.io/>). The Bayesian flow diagram is plotted by PGF/TikZ in LaTeX (<https://github.com/pgf-tikz>). The figures in supplementary text are plotted in R (Figs. S1 to S264), and the rest are plotted in Veusz. All the computations are handled in Google Colab.

Data availability

The complete dataset, codes employed for generating the outcomes, and the results of this research can be accessed at https://github.com/zboraon/bayesian_temperature_reconstruction.

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Supplemental material

Supplemental material for this article is available online.

Note

1. We use 'yrs BP' instead of 'years before present', where present corresponds to 1950 CE.

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